Code Coverage Report

Smart Complaint Management System · Team 5D
Statement coverage - Part 3

Version	1.0
Date of execution	12 Nov 2026, 15:45
Commit tested	66d1420
Branch	main
Test framework	pytest, pytest-cov (coverage.py), pytest-html
Overall result	114 of 114 automated tests passed

1. Summary

Statement coverage across the application code is 94.9% (1036 of 1092 statements executed). The figure combines all 114 automated tests (unit and system). Measured with coverage.py through pytest-cov; test files and scripts are excluded (.coveragerc), so the number reflects only product code.

2. Coverage by file

File	Statements	Missed	Covered %	Missing lines
app.py	37	5	86.5	31, 35, 39, 47-48
nlp/__init__.py	0	0	100.0	-
nlp/categorizer.py	15	0	100.0	-
nlp/preprocess.py	7	0	100.0	-
nlp/train_model.py	21	9	57.1	18-20, 24-27, 31-32
routes/__init__.py	0	0	100.0	-
routes/auth_routes.py	38	2	94.7	47-48
routes/complaint_routes.py	123	6	95.1	39, 103, 108, 120, 146, 166
routes/dashboard_routes.py	28	0	100.0	-
routes/escalation_routes.py	28	0	100.0	-
routes/evidence_routes.py	46	3	93.5	40, 42, 51
routes/notification_routes.py	34	0	100.0	-
routes/resolution_routes.py	38	2	94.7	48, 50
routes/sla_routes.py	52	9	82.7	26, 51, 64-70
services/__init__.py	0	0	100.0	-
services/analytics.py	31	1	96.8	18
services/auth.py	91	6	93.4	45, 52, 58, 60, 106, 134
services/complaints.py	50	1	98.0	36
services/escalation.py	61	2	96.7	27, 85
services/evidence.py	42	1	97.6	30
services/notifications.py	150	3	98.0	61, 117, 134
services/resolution.py	51	0	100.0	-
services/routing.py	17	0	100.0	-
services/sla.py	80	2	97.5	32, 107
services/status.py	30	2	93.3	37, 55
services/ticket_id.py	22	2	90.9	33, 36
TOTAL	1092	56	94.9	

3. Analysis of uncovered code

• nlp/train_model.py (lowest, about 57%): the training routine and its __main__ block. CI trains the model once (python -m nlp.train_model) before the tests, and inside the run the saved model is loaded. Model quality is checked by

test_categorizer instead of retraining in every test.

• app.py: the HTML page routes (/, /track, /login) and the __main__ block that starts the server and the SLA thread. These are exercised in manual browser testing, not by pytest.

• routes/sla_routes.py: GET /sla/rules, the agent-only filter in /sla/alerts, and the 60-second background monitor (start_monitor). The monitor's work, run_check, is covered through POST /sla/check.

• routes/complaint_routes.py: GET /categories, and the error exits of category override (invalid category, no permission, unknown complaint) and of GET / PATCH on an unknown complaint ID.

• services/auth.py: create_user validation messages for an empty name, an unknown role and an agent with no department; a token for an account that was deleted or deactivated; seeding the first admin from ADMIN_EMAIL / ADMIN_PASSWORD.

• Remaining single lines are similar error exits: 404/403 on listing notes and evidence, an unknown status name, the real SMTP login, and the ticket-ID guards for more than 99,999 tickets a day or a duplicate.

• The gaps are start-up code and rare error paths. Every requirement that is implemented has its main path covered.

4. Limitations of this metric

Statement coverage shows which lines ran, not whether their results were checked; it is backed here by assertions in every test. It cannot show missing features (TC06, TC14, TC15, TC19 are not implemented, so there is no code to cover), and it says nothing about the templates and JavaScript, which are exercised only in manual browser testing. Branch coverage, in the companion report, is the stricter measure.

5. Evidence

docs/reports/coverage-html/index.html (browse line by line), coverage.json and coverage.xml (Cobertura format).